

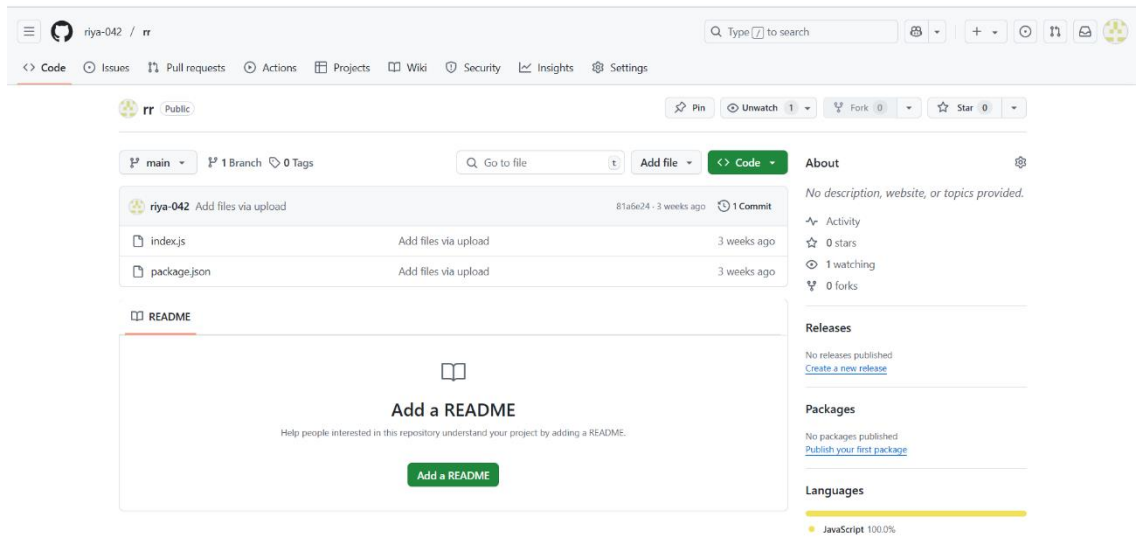
Assignment No: 09

Deploy a project from GitHub to EC2.

Solution:

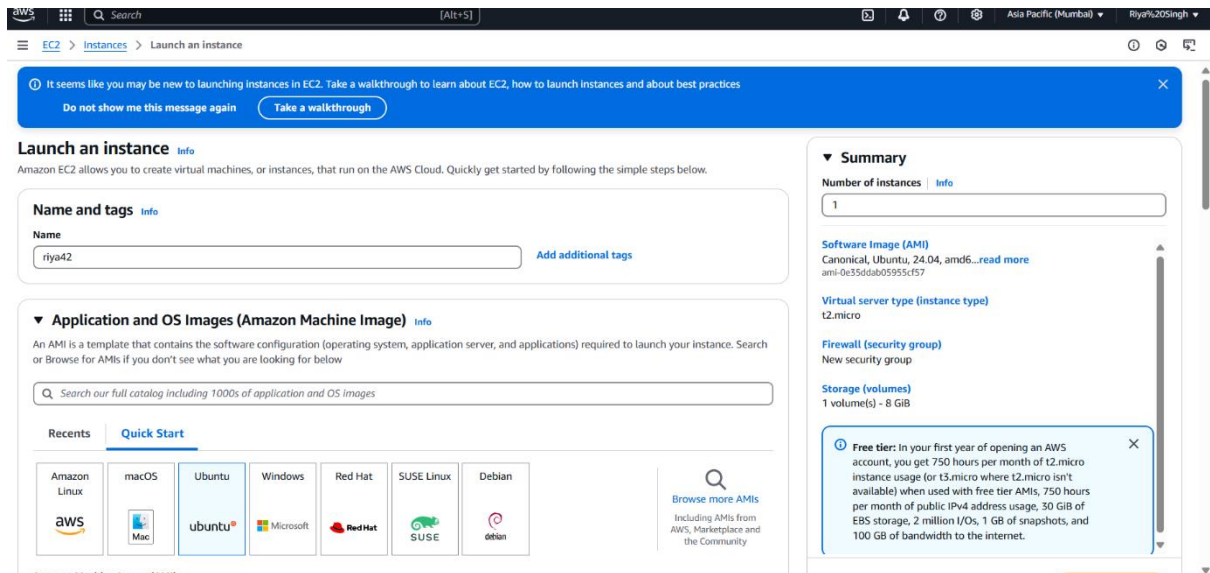
Step 1:

Upload required files to GitHub

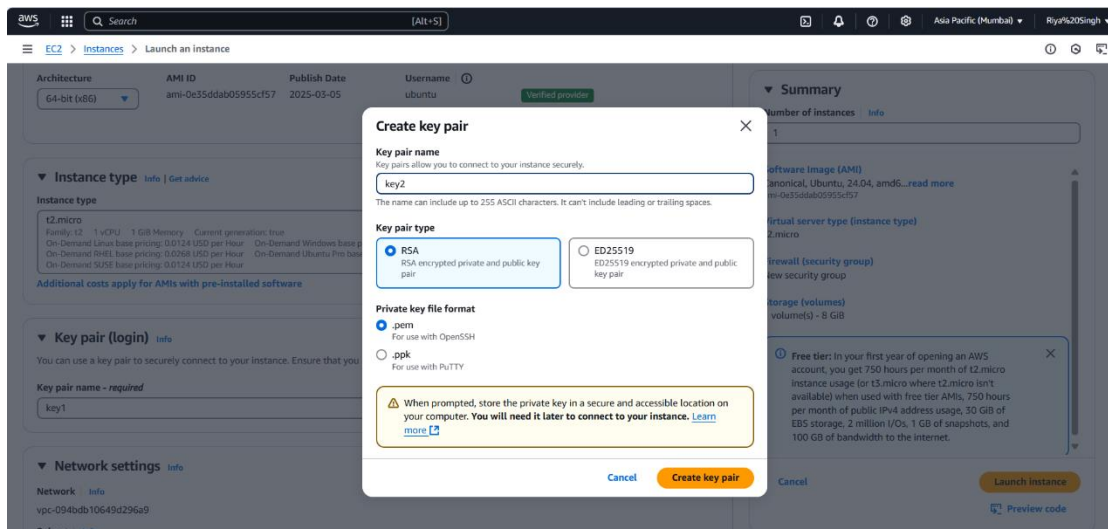


Step 2:

Log into AWS and open EC2. Click on launch instance. Give a name to the instance and select the operating system as Ubuntu.

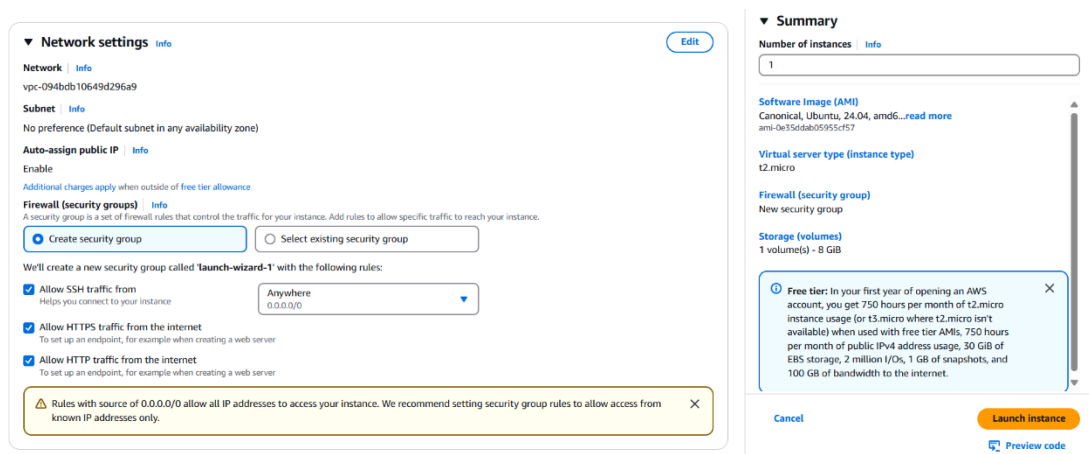


Step 3: Create Key Pair.

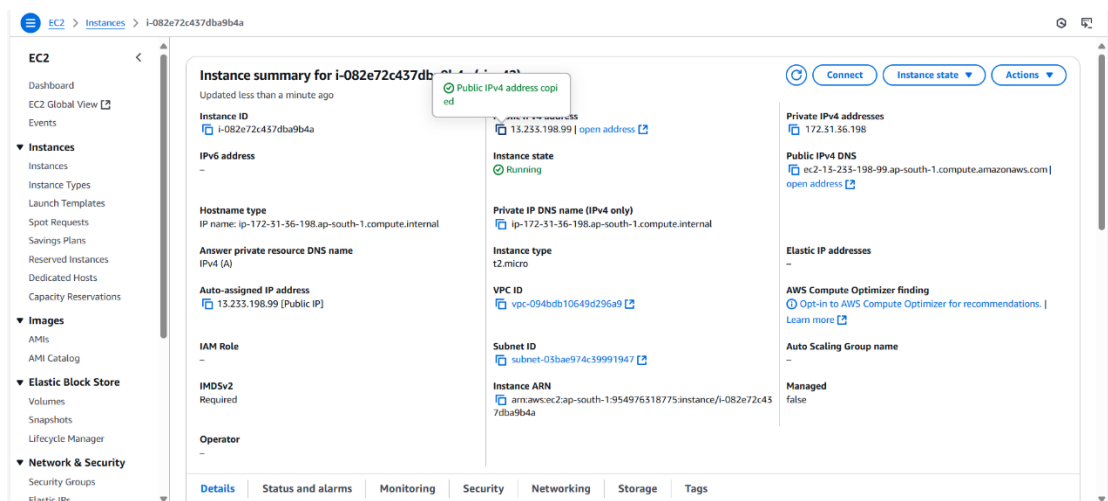


Step 4:

Check the 3 boxes to allow SSH, HTTP and HTTPS traffics and click launch instance.

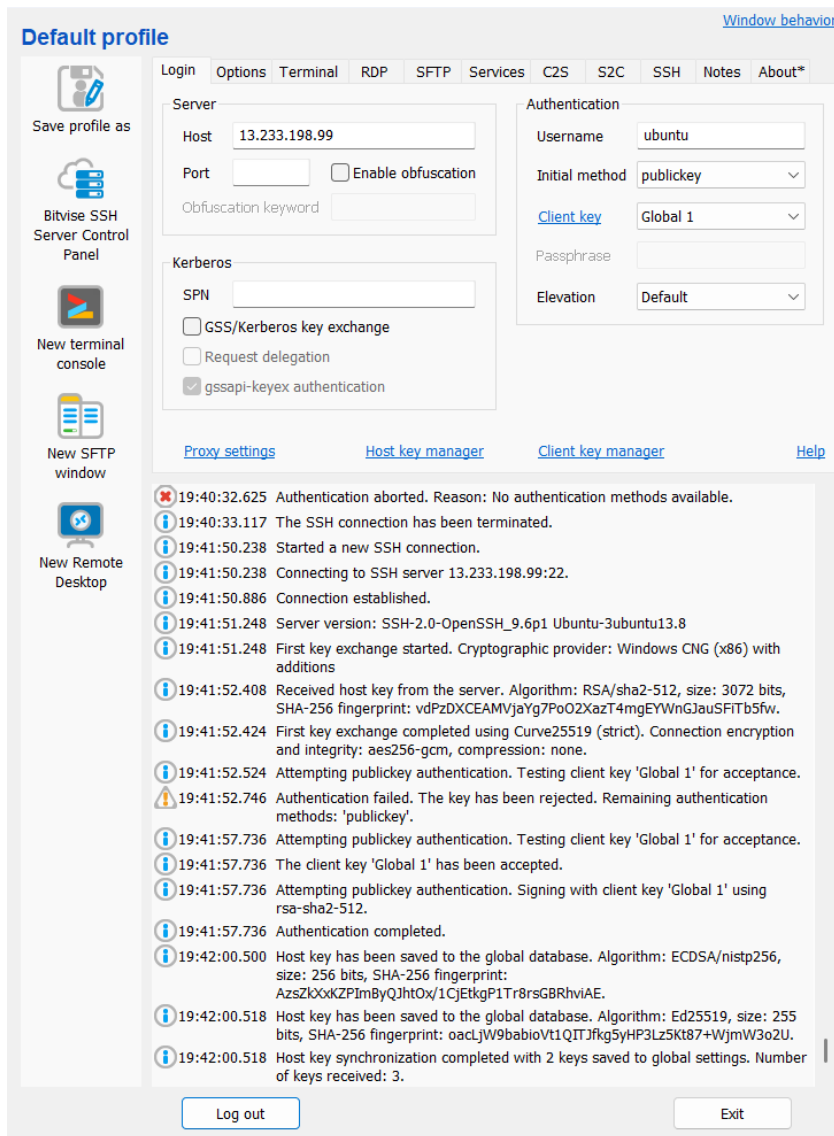


Step 5: Open instances and click of the instance Id of the instance. Copy the IPv4 address.



Step 6:

Open Bitwise SSH Client and paste the IP address in host and set up authentication using the key pair then login and after logging in open new terminal window.



Step 7:

Then Execute command `sudo apt-get update` and `sudo apt-get upgrade` in the Bitvise SSH Client Terminal.

```
ubuntu@ip-172-31-24-135:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-24-135:~$ sudo apt-get update

Fetched 33.3 MB in 7s (4447 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-24-135:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
```

Step 8:

Now Execute `sudo apt -get install nginx` Cmd for setting up the web server.

```
ubuntu@ip-172-31-36-198:~$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
```

Step 9:

After that, then open the directory that is cloned using command `cd` name of the directory. Then install NodeJS using commands `curl`:

https://deb.nodesource.com/setup_18.x | `sudo -E bash -` and `sudo apt-get install nodejs`.

```
ubuntu@ip-172-31-24-135:~$ curl -SL https://deb.nodesource.com/setup_18.x | sudo -E bash -
Get:6 https://deb.nodesource.com/node_18.x nodistro/main amd64 Packages [11.5 kB]
Fetched 23.7 kB in 1s (36.8 kB/s)
Reading package lists... Done
2025-04-10 17:51:33 - Repository configured successfully.
2025-04-10 17:51:33 - To install Node.js, run: apt-get install nodejs -y
2025-04-10 17:51:33 - You can use N|solid Runtime as a node.js alternative
2025-04-10 17:51:33 - To install N|solid Runtime, run: apt-get install nsolid -y
ubuntu@ip-172-31-24-135:~$
```

```
ubuntu@ip-172-31-24-135:~$ sudo apt install nodejs
Reading package lists... Done
```

Step 10: Now clone the GitHub repository from GitHub using command `git clone` paste the link of the repository

```
ubuntu@ip-172-31-36-198:~$ git clone https://github.com/riya-042/rr.git
Cloning into 'rr'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (4/4), done.
```

```
ubuntu@ip-172-31-40-111:~$ cd rr
ubuntu@ip-172-31-40-111:~/rr$ ls
index.js  package.json
ubuntu@ip-172-31-40-111:~/rr$
```

Step 11: After installing nodejs use `npm install` command to install required dependencies.

```
ubuntu@ip-172-31-36-198:~/rr$ npm install

added 227 packages, and audited 228 packages in 16s

25 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.3.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.3.0
npm notice To update run: npm install -g npm@11.3.0
npm notice
```

Step 12: Now open AWS and open the current instance then scroll down and go to security Section. Then click on the link in security groups then click on edit inbound rules.

The screenshot shows the AWS Management Console for the security group 'sg-0d44a0acec4d8305e - launch-wizard-1'. The 'Inbound rules' tab is selected, showing a table with 3 rules:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-04dfc388703b838ef	IPv4	SSH	TCP	22	0.0.0.0/0
-	sgr-0ee0bd1c135190494	IPv4	HTTP	TCP	80	0.0.0.0/0
-	sgr-0682811402b246cec	IPv4	HTTPS	TCP	443	0.0.0.0/0

Step 13: Click on add rule and select the type of new rule custom TCP Port range 4000 source 0.0.0.0 or anywhere then click on save rule.

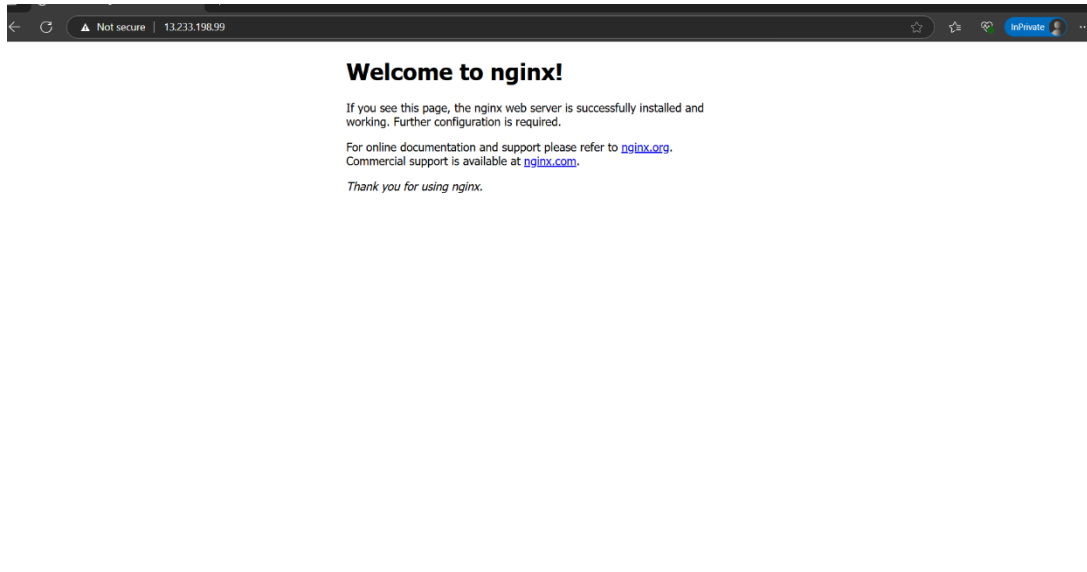
The screenshot shows the AWS Management Console after adding a new rule. A green notification banner at the top states: 'Inbound security group rules successfully modified on security group [sg-0d44a0acec4d8305e | launch-wizard-1]'. The 'Inbound rules' tab now shows 4 rules:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-04dfc388703b838ef	IPv4	SSH	TCP	22	0.0.0.0/0
-	sgr-0ee0bd1c135190494	IPv4	HTTP	TCP	80	0.0.0.0/0
-	sgr-0682811402b246cec	IPv4	HTTPS	TCP	443	0.0.0.0/0
-	sgr-05d3a244ad40158d1	IPv4	Custom TCP	TCP	4000	0.0.0.0/0

Step 14: Open the Bitwise SSH Client terminal console and start the server using command `node index.js`

```
ubuntu@ip-172-31-36-198:~/rr$ node index.js
Started server
```

Step 15: Now copy the IPv4 address of the instance and paste it on web browser.



Step 16:

Then, put the port like this IP address

